GazeCastGrasp: MR-Enabled Eye-Gaze Interaction for Assistive Robotic Manipulation

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ABSTRACT

We present *GazeCastGrasp*, a gaze-guided robotic manipulation system designed to enhance the intuitiveness and accessibility of assistive manipulators for individuals with mobility impairments. Traditional joystick- or keyboard-based interfaces require high precision and external reference frames, imposing heavy cognitive and physical demands. To address this, *GazeCastGrasp* leverages a Mixed Reality platform with integrated eye tracking to directly map visual attention to robotic actions. The system overlays augmented visual cues in the user's field of view and incorporates real-time gaze tracking, semantic alignment between user and robot perspectives, and progress feedback to enable natural gaze-to-grasp control. To validate the system design, we conducted a controlled user study comparing *GazeCastGrasp* with keyboard input in kitchen object retrieval tasks. Results showed that participants achieved a higher task success rate, reported greater usability and intuitiveness, and experienced lower workload with *GazeCastGrasp*. Overall, these findings demonstrate that MR-enabled gaze interaction can provide a more natural, accessible, and efficient modality for assistive robotic manipulation.

KEYWORDS

Eye-Gaze Interaction, Mixed Reality, Human-Robot Interaction, Assistive Manipulators

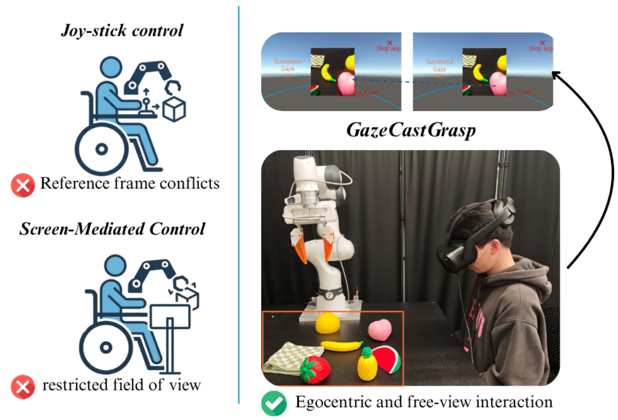

Figure 1: Motivation: natural and intuitive interaction using an assistive robotic manipulator remains a challenge.

1 INTRODUCTION

Robotic arms are increasingly demonstrating their potential to support everyday tasks, ranging from retrieving objects on tabletops to providing household assistance, and hold particular promise for individuals with limited mobility [17]. However, achieving natural and intuitive interaction using robots remains a central challenge. Most assistive manipulators rely on joystick- or keyboard-based interfaces, which not only demand precise hand movements and sustained attention but also require users to operate from external reference frames better suited to the robot's control logic [24]. This mismatch between user intention and robotic control significantly undermines usability and hinders adoption. To address these issues, researchers in Human–Robot Interaction (HRI) have explored alternative modalities such as gesture recognition, adaptive control schemes, and vision-based interfaces[13][19][26]. While these approaches extend accessibility to some extent, they often introduce trade-offs such as fixed viewpoints and constrained physical setups for cameras [26].

Among alternative modalities, gaze-based interaction has emerged as a particularly promising approach[26]. Eye movements are effortless, rapid, and naturally coupled with human intention [8]: when people fixate on an object, it often reflects their desire to interact with it. Unlike gestures or external control devices, gaze does not require additional physical effort or reliance on external reference frames, thereby offering a more direct mapping between intention and action. Prior work has demonstrated the feasibility of gaze-driven human–robot interaction for assistive manipulation and daily living support [25][6]. However, most existing systems relied on external screens or bulky head-tracking setups, which limited immersion and reduced intuitiveness. Meanwhile, these solutions do provide a user-friendly interface, limiting their applicability to real-life tasks.

To overcome these limitations, we present the *GazeCastGrasp* system, which enables users to directly control a robotic arm through their own gaze in a Mixed Reality (MR) environment. By wearing an MR headset, users can overlay virtual cues onto the real world. Instead of requiring the target object to be captured from the robot's perspective for secondary observation before grasping, they can simply fixate on the object they wish to grasp, eliminating the need for secondary observation from the robot's perspective. The robot then executes the corresponding action (shown in Figure 1), with progress feedback and control options integrated into the system..

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The system integrates user-centered gaze tracking, real-time object recognition, and semantic mapping between user and robot perspectives, achieving a seamless transition from visual attention to physical manipulation. With this system, we hypothesize that the gaze-based interface would yield higher usability, task success, trust, and user experience, while reducing workload.

Our contributions are threefold:

- We present a novel gaze-guided interaction system that leverages MR to align the user perspective with robotic perception.
- We implement a lightweight system architecture that combines gaze fixation, object recognition, and intent confirmation, enabling robust hands-free manipulation.
- We conduct an empirical evaluation showing that this approach improves intuitiveness, reduces latency, and achieves reliable grasping performance in real-world scenarios.

Together, these contributions show how gaze can serve as a natural and accessible control modality, paving the way toward more intuitive, user-centered assistive robotic systems.

2 RELATED WORK

2.1 Gaze Fixation

Gaze fixation, defined as the sustained visual attention directed at a particular object or spatial location, serves as a crucial indicator of both cognitive processing and social intention [3]. In instructional contexts, teacher gaze patterns have been shown to reflect specific pedagogical goals: fixating on students is often used to guide attention, on instructional content to emphasize key information, and on blank spaces to indicate cognitive effort, such as planning or hesitation [3]. Beyond temporal features, gaze fixation is also crucial to semantic inference and understanding of intention. Neurophysiological evidence indicates that observing another's gaze toward an object activates brain regions similar to those triggered by observing goal-directed actions, such as reaching or grasping [6]. These patterns suggest that both the duration and spatial target of a fixation can reveal an individual's cognitive load and communicative intent.

Fixation has also been shown to strongly correlate with attention allocation and cognitive effort. For example, in natural tasks, individuals tend to fixate on objects before acting on them, and shift gaze only once the action is nearly complete[6]. Moreover, gaze aversion—momentarily looking away—is often employed to regulate cognitive processing or reduce interpersonal stress, depending on the duration and context of the fixation[6]. As such, gaze as an intention proxy has been supported by numerous prior works.

2.2 Gaze-based interaction

Early research in HRI established the theoretical and methodological foundation for safe, efficient, and intuitive collaboration between humans and robots. In this emerging field, gaze behavior has been regarded as an important modality for conveying attention and intention. The study by Huang et al. [10] was one of the pioneering works in this direction, proposing an anticipatory manipulator control algorithm using real-time gaze tracking as an intention proxy, thereby enabling the robot to plan actions in collaborative tasks. The system combined eye gaze, speech input, and motion planning, and the experimental results demonstrated that gaze fixation patterns could reveal users' future goals and significantly improve the fluency and naturalness of interaction. With a similar approach, Aronson and colleagues [2] proposed a gaze-based hidden Markov model to infer the user's intention of interacting with objects and inform the control of the robotic arm. Their study demonstrated that gaze-based predictions enhance prediction quality and improve the performance of assistive robot arms.

Further to the gaze fixation events, the duration of fixation also plays an equally critical role in how the user's intent is interpreted. Short fixations (less than 1 second) are typically perceived as unintentional or non-interactive, whereas prolonged fixations (exceeding 2 seconds) are more likely interpreted as signals of interest, inquiry, or social engagement [14].

To address usability concerns, Namnakani et al. (2023) compared three widely used gaze interaction techniques under both stationary and walking conditions: Dwell Time, Pursuits, and Gaze Gestures. Their results showed that Pursuits achieved the fastest performance with a large number of targets, Dwell Time was more robust during walking, and Gaze Gestures offered the highest accuracy but at the cost of slower speed and increased cognitive demand[18]. In parallel, Narkar et al. (2024) proposed the GazeIntent system, which dynamically adjusts dwell-time thresholds through real-time intent modeling based on machine learning, achieving an F1-score of 0.94 and being preferred by 6% of returning users[20]. These pioneering works demonstrate successful practices for leveraging gaze in intent prediction.

More recently, Tokmurziyev et al. (2025) introduced GazeGrasp, a system that integrates eye tracking with YOLOv8-based object detection and introduces a magnetic snapping feature to reduce precision requirements and improve task completion time by 31%[26]. This study demonstrates the direct benefit of leveraging gaze patterns in assistive manipulators. However, GazeGrasp system relies on an external screen, which forces the user to attend to the objects on the screen rather than real-world objects, resulting in a restricted field of view and additional attentional shifts. These limitations highlight the need for immersive interfaces, such as MR, that can directly couple gaze input with real-world interaction to achieve more natural, efficient, and intuitive control.

2.3 MR interfaces

Mixed Reality (MR) has increasingly emerged as a powerful paradigm for enhancing human–robot collaboration. By providing immersive and context-aware visualizations, MR enables more natural interactions and effectively bridges the gap between operators and complex robotic systems. Walker et al. [29] conducted a comprehensive survey of Virtual, Augmented, and Mixed Reality in Human–Robot Interaction (VAM-HRI) and proposed a taxonomic framework to unify terminology and design elements across VR/AR/MR systems. Their work emphasized that MR interfaces can improve robot intent communication, enhance situational awareness, and facilitate naturalistic human control, while also supporting novel experimental methodologies and providing a conceptual foundation for advanced teleoperation systems. Building on this foundation, Ai et al. [1] introduced MR into the medical domain, presenting a HoloLens-based surgical robotics interface to replace traditional teleoperation consoles. By integrating hand gestures, head tracking, and speech commands, their system enabled mobile, sterile, and immersive surgical operations, demonstrating the transformative potential of MR for high-precision robotic tasks.

2.4 Assistive manipulation system

Assistive manipulation systems, as a key component of assistive technology, are designed to support individuals with physical impairments in performing activities of daily living (ADLs) and work-related tasks through robotic arms and other intelligent devices. Depending on the interaction context and application scenario, these systems are generally categorized into two main types: teleoperation and proximal interaction[24].

Recent studies show that despite decades of development, teleoperation remains largely confined to expert users and extreme domains such as disaster response, space exploration, and industrial inspection [5]. Rea and Seo [22] argue that its persistent challenges stem not only from technical limitations but also from the lack of user-centered design, with situation awareness and robot control continuing to impose high cognitive demands on operators. While teleoperation has proven effective for remote control, it generally assumes a spatial separation between the operator and the robot—conditions under which our approach of "directly fixating on an object with the eyes" is not feasible.

In contrast, assistive manipulator operations are primarily used in close-proximity interactions, where the user and the robot share the same physical space. Within this context, assistive manipulator operations can be viewed as a short-range extension of teleoperation, specifically designed for assistive grasping tasks. In addition to various input modalities explored in prior research, joystick-based interfaces remain a prevalent solution for assistive robotic manipulation, particularly in remote or indirect control scenarios. For example, Rulik et al. developed a wheelchair-mounted 6-DoF assistive robotic system that can be operated using either a finger joystick or a chin joystick, enabling users with different levels of upper-limb function to perform daily living tasks such as picking up objects from the floor, table, or shelves with high success rates and low latency[25] [23]. This work highlights the practicality of joystick-based control for long-range teleoperation, contrasting with our gaze-based approach, which is better suited for close-range, direct interaction.

3 SYSTEM DESIGN

3.1 Overview

Most gaze-based robotic systems rely on images captured from the robot's perspective or a relatively fixed head position of users [26]. While effective for algorithmic control, these setups distance the user from the physical task, forcing them to operate through an indirect viewpoint or specialized position. Our goal was to design a system that feels natural from the user's own perspective: looking at an object and having the robot act on it without additional translation steps.

We built a portable and easy-to-operate *GazeCastGrasp* system that combines a wearable Mixed Reality (MR) headset with a collaborative robotic arm to achieve this direct connection. Users observe the real environment through the headset with augmented visual cues and feedback overlaid. When a user fixates on a target, the system simultaneously captures the gaze coordinates and the scene image. Subsequently, the fixated target information is detected through a pretrained vision model. Then, the system transmits the fixated target data to the robotic control hub via the TCP communication protocol, where the ROS Master subscribes to incoming data. This enables the system to identify the object corresponding to the user's intention and send a grasping command to the robot.

This design ensures that users interact in an egocentric manner, choosing targets as they appear in their natural field of view, while the robot interprets and executes the action reliably. In doing so, *GazeCastGrasp* bridges immersive user experience with practical robotic manipulation, offering a hands-free workflow that aligns closely with intuitive human behavior. Packaging the system as an Android application provides access to the MR device's system panel, enabling program interruption through a button press. This provides a basic safeguard and highlights the potential for future integration of more seamless cancel or override mechanisms.

3.2 System integration

The proposed *GazeCastGrasp* system integrates multiple modules across the perception, interaction, and control layers to enable gaze-driven robotic grasping in an MR environment. The overall system architecture is illustrated in Figure 2.

In the perception layer, the MR headset continuously captures the user's gaze vector and scene images through its embedded eye tracker and passthrough cameras on the HTC Vive Focus Vision [9]. The MR environment provides an egocentric visual context, allowing users to naturally interact with physical objects through gaze in real-world settings.

After perception, data are passed to the processing layer. The gaze coordinates and image streams are transmitted in real time to a Windows workstation, where they are fused with YOLOv8-based object detection results to achieve target recognition and gaze matching. The system overlays bounding boxes and object labels on the detected targets, providing visual feedback and enhancing situational awareness.

In the control layer, the confirmed target label and command are transmitted via the TCP protocol to an Ubuntu system. The ROS Master listens for incoming messages, parses the object label, and commands the Franka Research 3 robotic arm to perform grasping and placement operations.

The task workflow begins when the user fixates on a target object for more than a threshold time t and manually confirms it via the MR interface. After image and gaze fixation are captured and processed through the perception and processing layers, the detected target label drives the robotic grasping model performing object retrieval. Once the object is successfully grasped and placed in the designated area, the task terminates automatically. Users can also manually trigger a reset command within the MR interface to end the session.

By seamlessly integrating MR visualization, gaze-based input, and robotic control, the system establishes a direct mapping from visual attention to action execution, expected to reduce cognitive load and enable a natural, hands-free human–robot collaboration experience.

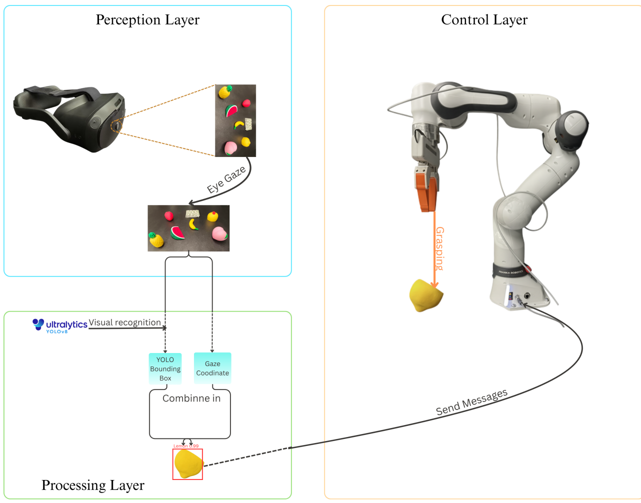

Figure 2: Overview of the GazeCastGrasp system

3.3 Gaze tracking

To establish the mapping between gaze targets and real-world objects, we design a workflow that maps eye gaze hit points from Unity's 3D coordinate space to 2D image space and associates them with YOLO-detected object labels. To acquire the eye gaze fixation data in the MR environment, we utilize the MR headset that activates the embedded eye tracker to capture the user's gaze direction in headset coordinates as a 3-D normal vector $\mathbf{n} \in \mathbb{R}^3$ (The z value remains unchanged). We then create a raycast that originates at the midpoint between the two eyeballs and points in the direction calculated by the average gaze directions $\overline{n}$ from the two eyeballs. The raycast projects through the virtual passthrough layer in the MR interface and collides with a collision plane precisely aligned with the passthrough layer. The hit point $\mathbf{x} \in \mathbb{R}^3$ is recorded corresponding to a screen pixel coordinate (x, y), where $x \in [0, W]$, $y \in [0, H]$, and H and W denote the MR screen height and width resolution, respectively. The colliding point (x, y) is recorded as a gaze fixation point at the passthrough image coordinate, as illustrated in Figure 3.

After the captured image is exported to the host computer, its pixel resolution on disk or during processing may differ from the original MR device resolution. To ensure coordinate consistency, we rescale the gaze point from the Unity screen space (x, y), defined under the MR device resolution $H \times W$, to the coordinate system $\mathbf{p} = (x', y')$ of the image on the PC with resolution $H' \times W'$. This transformation is computed as:

$$x' = \frac{x}{W} \cdot W', \quad y' = \frac{y}{H} \cdot H'$$

Grasp intention is recognized only when two requirements are satisfied: (i) binocular fixation on the same object for $\geq t$, and (ii) the participant manually confirms grasping in the MR interface. The captured image provided a reference for target selection.

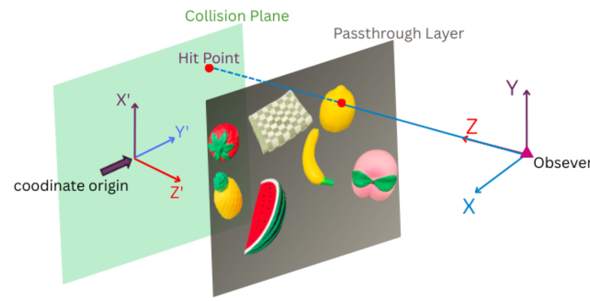

Figure 3: Gaze tracking: MR layer setup and projection

3.4 Object detection and recognition

After acquiring the transformed pixel coordinate of the gaze fixation, the system determines the object that the user fixates on. To enable object detection and recognition in the system, we employed YOLOv8[11] as the object detection framework. In a simulated kitchen environment, we captured images of seven categories of objects commonly seen in kitchens and relevant to assistive manipulator retrieval tasks (e.g., lemon, peach, towel, etc.), with a total of 135 images collected across all categories, constructing a custom dataset with 945 labeled instances and an object label set $\mathbf{L}$.

The dataset contains multiple images captured from different viewpoints, corresponding to the free-viewing conditions of the system and enhancing the model's detection robustness across perspectives. Each image was manually annotated using the labelImg tool [27], assigning bounding boxes and class labels to target objects. Once trained, the model was integrated into the system to provide object localization within the environment, supporting gaze-based target selection and robotic manipulation. Note that YOLO is sufficient for this pilot-scale study, while other robust computer vision models (e.g., Vision Language Models or Segment Anything [16]) can be deployed tailored to specific needs.

Each object detected by the trained object recognition model is represented by a rectangular bounding box, defined by its top-left and bottom-right pixel coordinates $[\mathbf{p}_1, \mathbf{p}_2]$.

To determine whether the user's gaze intersects any detected object, we check if the transformed gaze pixel coordinate (x', y') lies within any bounding box:

$$x_2 \geq x' \geq x_1, \quad y_2 \geq y' \geq y_1$$

If the condition is satisfied, the corresponding label $L_i \in \mathbf{L}$ is assigned as the semantic identity of the gaze target:

$$L_{\text{gaze}} = L_i$$

This procedure enables semantic grounding of gaze in a real-world MR environment by bridg ing gaze tracking, coordinate transformation, and object detection.

3.5 Robot communication and grasping

In the Windows workstation, after receiving the information transmitted from the VR device, the system first integrates the data and inputs it into the pretrained YOLO model for object recognition. Upon successful recognition, the system outputs the corresponding object label (labeled identifier). The processed results are then transmitted to the Ubuntu system via TCP protocol, where the ROS Master continuously listens within the local network for incoming messages (e.g., Message events). Each message contains key fields, including an integer-based object identifier and the bounding box coordinates in the camera frame. Finally, the system transforms this information into the robotic manipulation control command for object retrieval.

In this system, the Franka Research 3 robotic arm executes all grasping and placement operations once a command is issued. The arm first moves to a pre-grasp position above the target, then descends vertically until the gripper contacts the object, and finally closes the gripper to secure it. After lifting the object, the arm places it into a designated area. This totally hands-free process ensures repeatable and fully automated interaction.

4 USER STUDY

4.1 Experimental Design

To validate the system design, we designed and performed an experiment to measure the performance as well as participant feedback. We designed the experiment under two grasping systems: gaze-driven grasping using *GazeCastGrasp*(Treatment condition) and keyboard-controlled grasping (Control condition). We implemented a keyboard-based control as the control condition because it is the most common and default control method for assistive manipulators. Specifically, seven keys on the keyboard were utilized to directly control the robot arm: six keys for forward and backward translational movements along the x, y, and z axes, one key for gripper control. Users in keyboard condition assume full responsibility for the grasp. In sum, in the treatment condition, participants fixated on an object for at least $t = 2$ s. With the MR panel visualizing that intention detected, the participants then manually confirmed the interaction intention through the MR controller, which triggered perception and processing layers and passed the data to the robot. In the control condition, participants manually controlled the end-effector along the $x/y/z$ axes using a keyboard interface and executed grasp operations. Together, these two conditions enable a direct comparison between an automatic, intention-driven approach and a manual, precision-driven baseline.

The experiment adopted a within-subjects design comparing two conditions as shown in Figure 4. To reduce temporal interaction and confounding learning effects among conditions, the order of conditions was randomized across participants.

In order to make the experiment closer to the real use scenarios of people with disabilities, we designed the experimental scene. We used seven categories of everyday objects to simulate the kitchen scene in daily lives, including strawberry, pineapple, peach, watermelon, banana, towel, and lemon, as shown in Figure 5. These objects were randomly positioned on a tabletop, which means for the same object, the position and posture of the object should not be the same in two consecutive experiments. At the same time, we asked the participants to randomly choose which object to grasp, and required that the object selected by the same participant under two conditions should not be the same, and that two consecutive participants under the same condition should not select the same object. In this way, we maximized the randomness and fairness of the experiment and eliminated the learning effect.

To more realistically simulate the grasping behavior of individuals with mobility impairments, participants were instructed to perform the task while seated in a chair with a restricted range of movement. At the same time, no constraints were imposed on head movement or gaze direction, allowing participants to freely adjust their viewpoint and head posture within a comfortable range to maintain naturalistic interaction and ecological validity.

4.2 Data collection

We collected both objective and subjective measures. The primary objective metric was the grasp success rate. A grasping trial is considered successful if the object is stably grasped and accurately placed in the target area. Failures, by contrast, include unsuccessful gripping, object slippage, or task timeout (>2 minutes).

To evaluate the subjective feedback of the system design, participants completed a standardized questionnaire covering four dimensions: Usability, Workload, User Experience, and Trust. Responses were collected using 7-point Likert scales (1 = very negative, 7 = very positive). Questions are shown as Table 1. Note that the scale at W1 was that less discomfort (more comfortable) was associated with higher values. In designing the questionnaire, we drew upon several validated scales to ensure the scientific rigor and comparability of the evaluation results. The usability dimension was informed by the Technology Acceptance Model (TAM) [4], which emphasizes perceived ease of use and perceived usefulness, effectively assessing users' understanding of and learning costs associated with a new system. The workload dimension was based on the NASA-TLX [7], a widely applied tool in human–computer interaction and cognitive workload research. The user experience dimension adopted the framework of the User Experience Questionnaire (UEQ) [12], which evaluates naturalness, intuitiveness, sense of control, and comfort, helping to reveal the differences in interaction experience between the two control modalities. Finally, the trust dimension referred to the Trust in Automated System Test (TOAST) [15], which examines users' understanding of specific aspects of system functionality, as well as the consistency and predictability of system performance, thereby capturing participants' trust in automated grasping.

Trials were excluded if the target object was outside the reachable workspace or if a hardware malfunction occurred. A physical emergency stop was available at all times, and participants were free to withdraw at any point. All participants provided informed consent, and only anonymized participant IDs and system telemetry were recorded; no personally identifying information was stored.

4.3 Experiment process

After setting up the experiment environment, we briefly introduced the experiment process and how to operate the system to the participants. Each participant completed two grasp-and-place trials

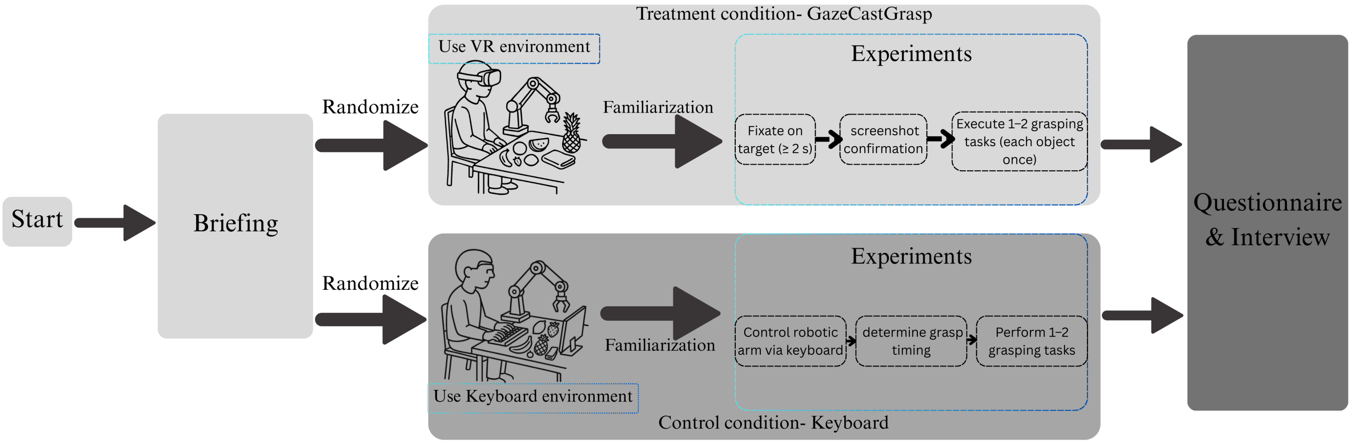

Figure 4: Experimental Pipeline

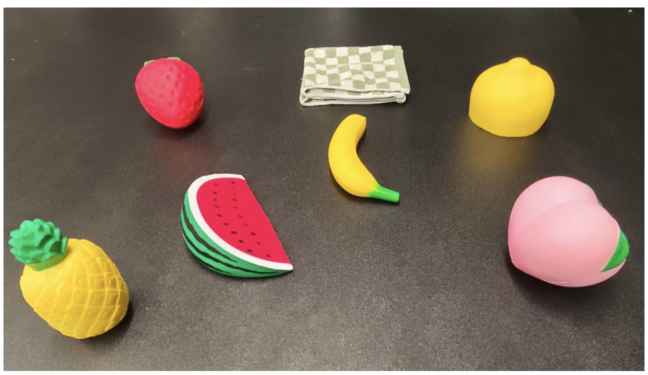

Figure 5: Experimental setup simulating a kitchen environment with graspable objects

Table 1: Questionnaire Design

Features	Question
Usability-U1	How easy is it to understand how the system works and operate it during tasks?
Usability-U2	How quickly do you feel you can master this system with practice?
Workload-W1	Do you feel any discomfort when using this system?
Workload-W2	How successful were you when completing the task?
User experience-UE1	How much did you feel in control while interacting with the system?
User experience-UE2	How natural and intuitive do you think this system is?
User experience-UE3	How willing are you to continue using this system if you were to retrieve objects using a robotic arm in the future?
Trust-T1	Do you trust the system can perform tasks accurately and safely?
Trust-T2	When an inaccuracy occurs, how much effort do you have to put into making it accurate?

under each condition. To mitigate fatigue and learning effects, a 30-second break was introduced between consecutive trials. The order of experimental conditions was counterbalanced across participants to minimize potential order effects.

The experimental procedure consisted of three phases: (1) Calibration and training: a polynomial gaze-tracking calibration built in MR was performed, followed by practice trials under both conditions to familiarize participants with the system and reduce the familiarization barrier. (2) Participants completed two randomized grasping tasks per condition to reduce the confounding order effect. (3) After each condition, participants completed a 7-point Likert questionnaire covering four dimensions: usability, workload, user experience, and trust, followed by an informal interview.

5 RESULTS

5.1 Participants

To ensure statistical power, we performed a power analysis on a pilot test with 5 participants using Pingouin [28]. With test power = 0.8 and significance level $\alpha = 0.05$, the highest required sample size yields 10.4 under question T1. As a result, we recruited a total of 14 participants. A post-hoc power analysis indicates that this sample size provides sufficient statistical power, with a minimum statistical power of 0.95, under the $\alpha = 0.05$ assumption.

All participants were graduate students with ages ranging from 23 to 28 years old (M = 24.9, SD = 1.68). None had prior experience with gaze-based robotic control, although several reported prior experiences with MR systems.

5.2 Performance

To further quantify the effectiveness and efficiency of the two control methods, we analyzed objective performance metrics across both experimental conditions. Under the Gaze-Based condition,

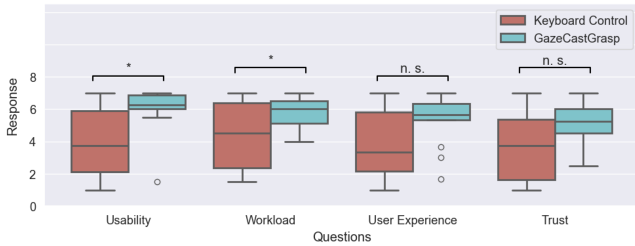

Figure 6: Questionnaire response results.

participants achieved a task success rate of 77.78%, whereas the Keyboard Control condition yielded a success rate of only 28.57%. This substantial difference indicates that gaze-driven interaction provides greater robustness and reliability in precise robotic manipulation. Post-experiment interviews further explained this gap: participants reported difficulty in mentally mapping their intended movements onto the robot's controls. For instance, when a participant intended to move the robot "left," the command did not always align intuitively with the robot's actual movement, due to mismatched reference frames between the human head orientation and the robot's base frame. In addition, participants highlighted a scale mismatch: achieving the fine-grained precision required near the object was particularly challenging with keyboard inputs, further limiting task success. Overall, these findings demonstrate that the gaze-based control method enables more reliable manipulation performance, supporting its potential application in real-world assistive scenarios.

5.3 Questionnaire Responses

To ensure consistent feature measurement, the responses under the same feature category are averaged, resulting in four features of interest: usability, workload, user experience, and trust. A statistical normality test (Anderson-Darling Normality Test [21]) was performed. The results indicated that the data deviated from a normal distribution at the 0.05 significance level. Given this violation of normality assumptions, we applied a non-parametric method (i.e., Wilcoxon signed-rank test) to test the null hypothesis that each pair of data was sampled from the same distribution. The statistical results are visualized in Figure 6.

Participants generally reported higher usability in *GazeCastGrasp* (p = 0.005, median = 6) compared with keyboard control (median = 3.75). According to the post-experiment interview, most participants under the gaze-based condition considered the system easy to understand and operate, with a short learning curve. Common descriptions included "intuitive" and "easy to learn." In contrast, keyboard control was perceived as more complex, with several participants noting that it required more time to master, primarily because of the mismatch in coordinate references between the user and the robotic arm.

For subjective workload (larger values denote lower perceived discomfort or load), the gaze-based condition again received a higher score compared to keyboard control (p = 0.050, median = 5.75 and 4.5, respectively). Although some participants reported experiencing eye fatigue during prolonged fixation, participants perceived

Table 2: Significance and Interpretation of Questionnaire Items

Question #	p-value	Interpretation
U1 (Q1)	0.005	Gaze scored higher, showing better usability.
U2 (Q2)	0.008	Gaze rated easier to use than keyboard.
W1 (Q3)	0.208	Discomfort level is similar between both conditions.
W2 (Q4)	0.019	User perceives using GazeCastGrasp more successful
UE1 (Q5)	0.193	The difference between levels of control is not significant
UE2 (Q6)	0.045	Gaze-based control is considered more intuitive
UE3 (Q7)	0.107	Preference is mixed
T1 (Q8)	0.066	The level of trust is mixed
T2 (Q9)	0.037	Gaze-based control is perceived easier for correction

significantly higher task success and lower discomfort/load under the gaze-based condition than under the keyboard condition.

User experience showed only marginal significance (p = 0.069). Although some participants reported that the gaze interface felt more natural, intuitive, and gave a stronger sense of agency, several responses suggest that gaze-based control is "less under control". This mixed response might suggest variations in users' previous experiences with MR systems. In addition, the willingness to continue using the system was significantly higher in the gaze-based condition compared to the keyboard condition. These results indicate that participants not only found gaze-based control more attractive but also expressed interest in adopting it for future robotic manipulation tasks.

Reported trust again showed only marginal significance (p = 0.054). During post-experiment interviews, most participants under the gaze-based condition expressed higher confidence, especially when the system reliably recognized targets. However, some participants raised concerns about potential "false triggers." In contrast, keyboard control was regarded as more transparent, providing participants with a stronger sense of safety during operation, albeit at a lower efficiency.

We further summarized the observed effects among all individual questions in Table 2. The trend generally aligns with a composed approach.

Overall, the quantitative results indicate that gaze-based control significantly outperformed keyboard control in terms of usability, workload, and task success; however, it also elicited mixed responses regarding trust and user experience, highlighting the trade-off between automation and trust.

6 DISCUSSION AND CONCLUSION

The primary research question addressed in this study was whether gaze-based control provides a more intuitive and efficient interaction modality for robotic arm manipulation compared to traditional keyboard input. Our initial hypothesis is that the gaze-based interface would yield higher usability, task success, user trust, user experience, and lower workload. The results partially support this hypothesis: gaze-based control showed clear advantages in performance, usability, and workload, but user experience and trust outcomes were more mixed.

Compared with keyboard input, gaze-based control demonstrated superior task performance, higher perceived usability, and lower reported workload (higher questionnaire response indicates lower workload). Participants consistently indicated that the system was easier to understand and quicker to master, suggesting that gaze interaction establishes a more natural and intuitive mapping between user intent and robotic action. This alignment appears to reduce cognitive demands during both the learning phase and real-time operation. The improvement in task success rates further supports this conclusion: task completion under the gaze-based condition reached 77.78%, nearly 50% higher than the 28.57% observed under keyboard control. These results highlight the robustness and precision of gaze-driven control in supporting complex manipulation tasks.

However, the outcomes for user experience and trust were less conclusive. While participants valued the intuitiveness of gaze-based interaction and its facilitation of quick corrections, statistical analysis revealed no significant difference in overall user experience or trust when compared with keyboard control. Questionnaire data revealed substantial variability: some participants valued the natural feel of gaze input, while others perceived it as offering less precise control than the keyboard, resulting in mixed levels of confidence and personal preference. Interviews further indicated that while gaze was intuitive for gross motions, it sometimes felt limiting for fine-grained manipulation. As such, the benefits of gaze-driven control appear to depend on individual preferences and expectations regarding precision and control.

In summary, *GazeCastGrasp* enhances efficiency, usability, and task success in robotic arm manipulation, making it a promising modality for intuitive human–robot interaction. Nevertheless, its impact on user trust and overall experience appears contingent on individual preferences and perceptions of control. These mixed outcomes suggest that while gaze-based interfaces can reduce workload and accelerate learning, additional refinements—such as combining gaze with secondary input modalities—may be necessary to achieve both high efficiency and high user confidence across diverse users.

Although this study has made positive progress in the field of gaze-based human–robot collaboration, several limitations remain. First, the controlled laboratory setup simulated a kitchen scenario but did not capture the complexity of real-world environments, where lighting variations, distractions, and multitasking may impact gaze-tracking accuracy and interaction performance. Field or semi-naturalistic studies are needed to assess ecological validity. Second, the study examined only short-term interactions, leaving long-term factors such as trust development, skill acquisition, and fatigue unaddressed. Longitudinal evaluations would provide insights into these dynamics and inform systems designed for sustained use. Finally, due to ethical considerations and the need for pilot evidence, participants in this study were healthy adults, while the system is primarily intended to support individuals with limited mobility. Feedback and performance may differ substantially in the target population because of physiological and psychological factors. Although we partially addressed this by constraining the motion area, future studies should recruit participants with mobility limitations to more accurately evaluate the system's effectiveness and usability.

Future research will focus on optimizing intelligent and adaptive systems. For example, we plan to explore personalized gaze thresholds tailored to individual differences and integrate micro eye and head movement data to enhance system sensitivity and responsiveness. In addition, machine learning–based dynamic calibration algorithms will be introduced to achieve more robust multi-user adaptability. Furthermore, we will integrate gaze intention with multi-modal interaction (e.g., speech and gestures) and shared autonomy mechanisms to improve system safety and interpretability. By incorporating path planning and multi-modal feedback, the system will be able to strengthen error recovery capabilities and reduce the likelihood of false triggers.

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REFERENCES

[1] Letian Ai, Peter Kazanzides, and Ehsan Azimi. 2024. Mixed Reality Based Teleoperation and Visualization of Surgical Robotics. *Healthcare Technology Letters* 11 (2024), 179–188. https://doi.org/10.1049/htl2.12079

[2] Reuben M Aronson, Nadia Almutlak, and Henny Admoni. 2021. Inferring goals with gaze during teleoperated manipulation. In *2021 IEEE/RSJ international conference on intelligent robots and systems (IROS)*. IEEE, 7307–7314.

[3] Akanksha Baranwal, Nitesh Goyal, and Rakesh Chaturvedi. 2021. Understanding Teacher Gaze Patterns and Student Engagement in the Classroom Using a Mixed-Methods Approach. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–14. https://doi.org/10.1145/3411764.3445390

[4] Fred D. Davis. 1989. Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Quarterly* 13, 3 (1989), 319–340. https://doi.org/10.2307/249008

[5] Anca D. Dragan and Siddhartha S. Srinivasa. 2012. Assistive Teleoperation for Manipulation Tasks. In *Proceedings of the 7th ACM/IEEE International Conference on Human-Robot Interaction (HRI '12)*. ACM, Boston, Massachusetts, USA, 123–124. https://doi.org/10.1145/2157689.2157716

[6] Matthew Goodwin, Allison M. Okamura, and Katherine J. Kuchenbecker. 2021. Social Eye Gaze in Human-Robot Interaction: A Review. *Frontiers in Robotics and AI* 8 (2021), 748–763. https://doi.org/10.3389/frobt.2021.748763

[7] Sandra G. Hart. 2006. NASA-Task Load Index (NASA-TLX); 20 Years Later. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, Vol. 50. SAGE Publications, San Francisco, CA, USA, 904–908. https://doi.org/10.1177/154193120605000909

[8] Mary Hayhoe and Dana Ballard. 2005. Eye movements in natural behavior. *Trends in Cognitive Sciences* 9, 4 (2005), 188–194. https://doi.org/10.1016/j.tics.2005.02.009

[9] HTC. 2025. *Vive Focus Vision.* https://www.vive.com/us/?cjdata=MXxOfDB8WXww&utm_medium=affiliate&utm_source=cj&cjevent=0864a4cc9e0211f081a500580a82b82a&wpsrc=cj&wpcn=affiliate&wpsnetn=cj Accessed 30 September 2025.
[10] Chien-Ming Huang, Maya Cakmak, and Bilge Mutlu. 2016. Anticipatory Robot Control for Efficient Human-Robot Collaboration. In *Proceedings of the 2016 ACM/IEEE International Conference on Human-Robot Interaction (HRI '16)*. IEEE Press, 83–90. https://doi.org/10.1109/HRI.2016.7451738
[11] Glenn Jocher, Ayush Chaurasia, and other contributors. 2023. *Ultralytics YOLOv8*. Accessed: 2025-07-09.
[12] Bettina Laugwitz, Theo Held, and Martin Schrepp. 2008. Construction and Evaluation of a User Experience Questionnaire. In *Proceedings of the 4th Symposium of the Workgroup Human-Computer Interaction and Usability Engineering of the Austrian Computer Society (USAB 2008)*. Springer, Graz, Austria, 63–76. https://doi.org/10.1007/978-3-540-89350-9_6
[13] Rung-Huei Liang and Ming Ouhyoung. 1996. A Real-time Continuous Gesture Recognition System for Sign Language. In *Proceedings of the ACM Symposium on Virtual Reality Software and Technology*. ACM, 59–66.
[14] François Loth, Christian Essel, and Elisabeth André. 2021. Robotic Gaze and Human Views: A Systematic Literature Review of Gaze in Human–Robot Interaction. *Frontiers in Robotics and AI* 8, 647693 (2021). https://doi.org/10.3389/frobt.2021.647693
[15] Bertram F. Malle and Daniel Ullman. 2018. A Multidimensional Conception and Measure of Human–Robot Trust. *HRI '18: Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (2018), 303–312. https://doi.org/10.1145/3171221.3171260
[16] Meta AI Research. 2023. Segment Anything Model (SAM). https://segment-anything.com/?utm_source=chatgpt.com. Accessed: 2025-09-30.
[17] Bruce Michalec. 2021. Assistive robotic arms could transform life for people with disabilities. *Science* 372, 6543 (2021), 404–407. https://doi.org/10.1126/science.aat8414
[18] Omar Namnakani, Ken Pfeuffer, Andreas Bulling, Benjamin Turner, and Eduardo Velloso. 2023. Comparing Dwell Time, Pursuits and Gaze Gestures for Gaze Interaction on Handheld Mobile Devices. In *Proceedings of the 2023 ACM Symposium on Eye Tracking Research & Applications (ETRA '23)*. ACM, Tubingen, Germany, 1–10. https://doi.org/10.1145/3588015.3588483
[19] Kumpati S. Narendra and Jeyendran Balakrishnan. 1997. Adaptive Control Using Multiple Models. *IEEE Trans. Automat. Control* 42, 2 (1997), 171–187.
[20] Anish S. Narkar, Jan J. Michalak, Candace E. Peacock, and Brendan David-John. 2024. GazeIntent: Adapting Dwell-Time Selection in VR Interaction with Real-Time Intent Modeling. In *Proceedings of the ACM Symposium on Eye Tracking Research & Applications (ETRA '24)*. ACM, Glasgow, UK, 226:1–226:18. https://doi.org/10.1145/3655600
[21] Nornadiah Mohd Razali, Yap Bee Wah, et al. 2011. Power comparisons of shapiro-wilk, kolmogorov-smirnov, lilliefors and anderson-darling tests. *Journal of statistical modeling and analytics* 2, 1 (2011), 21–33.
[22] Daniel J. Rea and Stela H. Seo. 2022. Still Not Solved: A Call for Renewed Focus on User-Centered Teleoperation Interfaces. *Frontiers in Robotics and AI* 9 (2022), 704225. https://doi.org/10.3389/frobt.2022.704225
[23] John Rulik, Bryan McClintock, and Thomas Smith. 2022. Control of a Wheelchair-Mounted 6DOF Assistive Robot With Chin and Finger Joysticks. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 1234–1241. https://doi.org/10.1109/ICRA.2022.1234567
[24] Linda Sørensen. 2025. *Assistive Humanoid Robots for Enhancing Independence, Autonomy, and the Ability to Engage in Meaningful Occupations: Perceptions and Experiences in Users with Physical Disabilities*. Ph. D. Dissertation. University of Agder, Faculty of Health and Sport Science, Kristiansand, Norway. Dissertation for the degree philosophiae doctor (Ph.D.).
[25] Md Samiul Haque Sunny, Md Ishrak Islam Zarif, Ivan Rulik, Javier Sanjuan, Mohammad Habibur Rahman, Sheikh Iqbal Ahamed, Inga Wang, Katie Schultz, and Brahim Brahmi. 2021. Eye-gaze control of a wheelchair mounted 6DOF assistive robot for activities of daily living. *Journal of NeuroEngineering and Rehabilitation* 18 (2021), 173.
[26] Issatay Tokmurziyev, Miguel Altamirano Cabrera, Luis Moreno, Muhammad Haris Khan, and Dzmitry Tsetserukou. 2025. GazeGrasp: DNN-Driven Robotic Grasping with Wearable Eye-Gaze Interface. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA '25)*. IEEE, Atlanta, GA, USA, 1–7. https://doi.org/10.1109/ICRA2025.123456
[27] Tzutalin. 2015. LabelImg. https://github.com/HumanSignal/labelImg. Accessed: 2025-09-18.
[28] Raphael Vallat. 2018. Pingouin: statistics in Python. *J. Open Source Softw.* 3, 31 (2018), 1026.
[29] Michael Walker, Thao Phung, Tathagata Chakraborti, Tom Williams, and Daniel Szafir. 2023. Virtual, Augmented, and Mixed Reality for Human-Robot Interaction: A Survey and Virtual Design Element Taxonomy. *ACM Transactions on Human-Robot Interaction* 12, 4 (2023), 43:1–43:39. https://doi.org/10.1145/3597623